Transport Reviews

ISSN: 0144-1647 (Print) 1464-5327 (Online) Journal homepage: www.tandfonline.com/journals/ttrv20

From aspiration to operation: ensuring equity in transportation

Anne Brown

To cite this article: Anne Brown (2022) From aspiration to operation: ensuring equity in transportation, Transport Reviews, 42:4, 409-414, DOI: 10.1080/01441647.2022.2064527

To link to this article: https://doi.org/10.1080/01441647.2022.2064527

Published online: 16 Apr 2022.

Submit your article to this journal

Article views: 3444

View related articles

View Crossmark data

Citing articles: 8 View citing articles

Full Terms & Conditions of access and use can be found at
https://www.tandfonline.com/action/journalInformation?journalCode=ttrv20

TRANSPORT REVIEWS
2022, VOL. 42, NO. 4, 409–414
https://doi.org/10.1080/01441647.2022.2064527

Routledge
Taylor & Francis Group

Check for updates

From aspiration to operation: ensuring equity in transportation

Anne Brown

School of Planning, Public Policy, and Management, University of Oregon, Eugene, OR, USA

Introduction

Transportation systems are, and have long been, profoundly inequitable. Private transportation services, such as e-scooters and microtransit, have evolved alongside cultural reckonings around systemic racism and unprecedented income inequality (among other inequities). In response, transportation professionals have amplified efforts to deliver equitable transportation. Research to date has examined transportation equity across a host of topics ranging from transportation and environmental justice (e.g. Schweitzer & Valenzuela, 2004), transportation finance (e.g. Taylor & Norton, 2009), unequal active transportation investments (e.g. Barajas, 2018), gender and safety (e.g. Lubitow, Carathers, Kelly, & Abelson, 2017), transit fare structures (e.g. Farber, Bartholomew, Li, Páez, & Habib, 2014) and discrimination in shared mobility (e.g. Moody, Middleton, & Zhao, 2019) to name just a few. In this editorial, I consider how transportation professionals can harness knowledge of transportation (in)equity to deploy equitable transportation programming. Specifically, I consider goal setting in transportation practice, centring communities in transportation planning, and the need for data to measure and evaluate progress.

Aiming for equity

No single definition of equitable transportation exists. In some ways this presents a challenge: should equity be evaluated as a process, opportunity, and/or outcome? Should it be measured across individuals, groups, or geographies (Taylor & Norton, 2009)? Yet its normative application (Lewis, MacKenzie, & Kaminsky, 2021) and definitional ambiguities also present opportunities for cities or programmes to define equity using the dimensions most relevant to their local context. Considering multiple dimensions of equity is imperative since defining equity goals as a single dimension may neglect intersectionality (e.g. gender *and* race) or dimensions of exclusion. For example, a city may promote opportunity equity by ensuring that shared micromobility (e.g. bikeshare or e-scooter) is deployed across all city neighbourhoods. Yet physical access to scooters or other forms of micromobility may be moot if these services remain inaccessible behind other barriers, such as the need for a credit card or smartphone in order to ride them.

Aiming for equity should begin with clearly defining programme goals. Transportation programmes or agencies often aim to advance multiple goals, from reducing climate

CONTACT Anne Brown abrown33@uoregon.edu
© 2022 Informa UK Limited, trading as Taylor & Francis Group

impacts to providing access to jobs; yet relatively few transit agencies or city micromobility contracts mention equity as a primary objective (Brown, Howell, & Creger, 2022; Taylor & Morris, 2015). Equitable outcomes may be aligned with other goals. For example, reducing environmental impacts may positively benefit peoples' health, particularly marginalised communities who bear disproportionate burdens of roadway emissions (Rowangould, 2013). Yet explicit equity goals remain imperative in order to incorporate equity considerations into all steps of programme delivery, from design, to data collection, to performance metrics, evaluation, and iteration.

Concrete goals may also help distinguish competing party interests. Groups historically marginalised by the transportation system are not monolithic; as such, policies that promote accessible and affordable transportation for one group may inadvertently undermine affordable access by another. For example, implementing cash-less transit fare systems could greatly benefit transit riders by expediting boardings and enabling fare capping strategies; yet removing cash payment options could also exclude riders without smartphones or bank accounts (Golub, MacArthur, Brown, & Brakewood, 2021). Such tensions raise important considerations such as who (or which groups or places) should be prioritised in transportation policy and planning. Transportation professionals must both weigh objectives and attempt to implement policies that address these intersectional and at times competing interests. Well-defined goals established upfront can guide transportation professionals in decision-making during later phases of programme design and evaluation.

Goal-making is a critical first step in transportation planning and policy to identify desired outcomes, target groups or geographies, and shape the programme design or delivery needed in order to achieve equity. When undertaken in collaboration with community stakeholders, goal setting can also help centre local needs within programme design.

Centring communities in transportation planning

Planners and policymakers have many tools at their disposal to increase transportation access and equitable outcomes. Sometimes, however, conversations focus on the tools themselves rather than how these tools will be used to improve equitable outcomes. For example, let's say a bikeshare pilot programme is underway and the city wants to ensure that vehicles are available in all neighbourhoods, particularly in areas that have historically been marginalised and underserved. On one hand, distributing vehicles across space is an important step in ensuring opportunity equity. On the other hand, expending efforts and focusing on deploying bikes could neglect broader transportation barriers that the community faces. How useful is bikeshare, for example, if the community is unwilling or unable to ride it given a dearth of locally supportive infrastructure? Best practice suggests that conducting community mobility needs assessments can help identify and prioritise local community transportation needs in order to deliver programmes and investments that have the greatest positive impact on resident mobility (Creger, Espino, & Sanchez, 2018). Conducting regular community needs assessments is time consuming and resource intensive. But needs assessments also allow cities to pivot quickly in a fast-changing transportation landscape. When e-scooters arrived on city streets, for example, cities faced mounting pressure to quickly enact regulatory

policies to shape fleet deployment and operations; with imminent needs, many cities implemented policies first and conducted community outreach second, which often precluded access for marginalised travellers and neighbourhoods. Although many cities successfully iterated programmes to be more inclusive following pilot evaluation and community dialogue (demonstrating the value of both pilots and programme iteration to achieve more equitable outcomes), an ongoing understanding of community needs can yield supportive programme design from the start (Brown et al., 2022).

Older, male, white homeowners are disproportionately engaged in the public participation process (Einstein, Palmer, & Glick, 2019). Engaging – and more importantly, empowering – marginalised groups in the transportation planning process are imperative to inform and shape interventions that would yield the greatest benefits to mobility and accessibility. Community engagement in transportation planning can take many forms, but ideally should reflect the voices and grant some degree of decision-making authority to residents. Engagement may mean partnering with or conducting outreach with and through community-based organisation that can mediate between transportation professionals and the community. Community groups can be important facilitators to help identify local mobility needs as well as assist in programme deployment and outreach such as distilling fare capping policies and how to take advantage of them to community members. Some cities also require community engagement events – and correspondingly track the number, types and locations of engagement efforts – as part of shared micromobility permits (Brown et al., 2022). Transportation professionals should likewise consider the ways in which digital tools – surveys, interactive tools, virtual meetings, etc. – can expand engagement beyond traditional in-person meetings. Research finds that while online methods do not ensure specific participatory outcomes, "online approaches, combined with traditional techniques, make pragmatic sense for scaling up participation" and can aid planners in reaching wider geographies and income levels (Griffin & Jiao, 2019). Likewise, a single engagement activity is unlikely to produce representative input from the community; instead, meeting community members where they are, and compensating them for their local expertise is imperative to support both community buy-in and prolonged engagement. Programme or agency budgets should account for engagement efforts that compensate participants or community-based organisations for their local expertise and perspectives.

In current public engagement models, public participation is often a single, defined step in the planning process. While ideally, public participation should be a continuous dialogue rather than a discrete step, at a minimum community engagement should happen *before* programme or project plans are solidified and empower the community to identify priorities and shape programmes to directly tackle the barriers they face and mobility needs they most need met. Finally, even the most empowering public participation models employed in practice, such as participatory budgeting, may be impeded by practical details such as compressed timelines and remain only as strong as the broad coalition that they bring to the table (Karner, Brower Brown, Marcantonio, & Alcorn, 2019).

To achieve transportation equity, we need data

Implementing policy in many ways is just the first step in a never-ending cycle of equitable transportation planning that rotates between goal setting, implementation,

evaluation, and iteration. Once a policy aimed to deliver equity is implemented, the next step must be to understand how well (or not well) the programme is meeting its objectives. Understanding the relative successes can help to both identify best practices moving forward, as well as how programme iteration may address its identified shortcomings.

Following programme implementation, cities or agencies should evaluate programmes or policies. One cannot understand where programmes succeeded and where changes should be made without first evaluating a programme. This means building data collection and evaluation into programme or pilot timelines, resources, and service contract agreements with private mobility companies, where applicable. The data that cities collect are critical to programme evaluation; each datapoint can answer some questions but not others. Required or collected data should align with programme objectives or key performance indicators. For transit, this may mean examining bus on time performance across space. For private shared micromobility companies (bikeshare, e-scooters), many cities require companies to provide publicly-accessible trip data that includes information on trip origin, destination, cost, and distance. These data can help answer important questions related to service deployment and use. However, trip-level data cannot answer all equity-related questions, particularly questions around user characteristics or trip purposes. An added challenge is that surveys needed to gather these data may prove prohibitively expensive and/or challenging for cities to administer. One strategy cities have implemented in order to balance data needs with limited resources is to require private transportation companies operating within their jurisdiction to shoulder survey deployment. Some cities request that operators provide disaggregated, anonymised demographic data. Others, such as Baltimore City Department of Transportation (BCDOT), require shared electric scooter companies to deploy an annual survey, written by agency staff, to all users. Surveys can help provide important data about users and user experiences that cannot be gleaned from trip data alone, and allow for qualitative responses to reflect user experiences.

Cities should use results from data analysis and evaluation to identify gaps in service or where equity requirements or programming may fall short. For example, just 43 people signed up for Portland's (Oregon) reduced fare scooter programme during its first pilot year; the second year reported similarly low enrolment rates at just 0.9% of users (PBOT, 2019, 2020). One possible reason could be residents' limited knowledge of the programme or may reflect barriers to enroll in the reduced fare programme. In Portland, for example, people using cash to buy credit for Spin Access (the reduced e-scooter fare programme) must visit the local Spin office, which is only open Mondays from 11am to 2pm (Spin, 2022). Such barriers can undermine programmatic requirements such as requiring a cash payment option, and preclude equity aspirations from being operationalised. This case also highlights how barriers to access are rarely singular; efforts to remove barriers must be ongoing and often require concerted and targeted efforts. Targeted efforts may include partnerships across organisations such as co-locating bikeshare stations and affordable housing and providing memberships to residents. Some transportation agencies also build in either incentives or deterrents into contracts with shared micromobility companies to promote programme equity, such as reducing vehicle fees in targeted neighbourhoods. Using data to identify gaps in service across rider groups (e.g. race,

income, gender) or geographies may help cities iterate programmes such as adding new components or partnering with new community-based organisations to target outreach efforts in select areas. While difficult to collect, understanding who does *not* use transportation services, be it transit or shared mobility, can likewise aid cities in identifying barriers to transportation.

Transportation planning and programmes can be designed with best practices learned from other cities. Yet in the ever-changing landscape of transportation, new untested ideas and modes will appear and policies will remain untested in new contexts. Setting clear objectives – including equity goals informed by community mobility needs assessments – matched by data collection and evaluation efforts can enable iteration to ensure that aspirations for transportation are operationalised to deliver affordable and accessible transportation options for all.

Disclosure statement

No potential conflict of interest was reported by the author(s).

ORCID

Anne Brown http://orcid.org/0000-0001-5009-8331

References

Barajas, J. M. (2018). Not all crashes are created equal: Associations between the built environment and disparities in bicycle collisions. *Journal of Transport and Land Use*, *11*(1), 865–882.

Brown, A., Howell, A., & Creger, H. (2022). *Mobility for the people: Equity in shared micromobility programs*. National Institute for Transportation and Communities.

Creger, H., Espino, J., & Sanchez, A. S. (2018). *Mobility equity framework*. Greenlining Institute. https://greenlining.org/wp-content/uploads/2019/01/MobilityEquityFramework_8.5(11_v_GLI_Print_Endnotes-march-2018.pdf

Einstein, K. L., Palmer, M., & Glick, D. M. (2019). Who participates in local government? Evidence from meeting minutes *Perspectives on Politics*, *17*(1), 28–46.

Farber, S., Bartholomew, K., Li, X., Páez, A., & Habib, K. M. N. (2014). Assessing social equity in distance based transit fares using a model of travel behavior. *Transportation Research Part A: Policy and Practice*, *67*, 291–303.

Golub, A., MacArthur, J., Brown, A., & Brakewood, C. (2021). *Applying an equity lens to automated payment solutions for public transportation*.

Griffin, G. P., & Jiao, J. (2019). The geography and equity of crowdsourced public participation for active transportation planning. *Transportation Research Record: Journal of the Transportation Research Board*, *2673*(1), 460–468.

Karner, A., Brower Brown, K., Marcantonio, R., & Alcorn, L. G. (2019). The view from the top of Arnstein's ladder: Participatory budgeting and the promise of community control. *Journal of the American Planning Association*, *85*(3), 236–254.

Lewis, E. O., MacKenzie, D., & Kaminsky, J. (2021). Exploring equity: How equity norms have been applied implicitly and explicitly in transportation research and practice. *Transportation Research Interdisciplinary Perspectives*, *9*, 100332.

Lubitow, A., Carathers, J., Kelly, M., & Abelson, M. (2017). Transmobilities: Mobility, harassment, and violence experienced by transgender and gender nonconforming public transit riders in Portland, Oregon. *Gender, Place & Culture*, *24*(10), 1398–1418.

Moody, J., Middleton, S., & Zhao, J. (2019). Rider-to-rider discriminatory attitudes and ridesharing behavior. *Transportation Research Part F: Traffic Psychology and Behaviour*, *62*, 258–273.

PBOT. (2019). 2018 *E-Scooter Findings Report*. https://www.portland.gov/sites/default/files/2020-04/pbot_e-scooter_01152019.pdf

PBOT. (2020). 2019 *E-Scooter Findings Report*. https://www.portland.gov/sites/default/files/2020/pbot_e-scooter_report_final.pdf

Rowangould, G. M. (2013). A census of the US near-roadway population: Public health and environmental justice considerations. *Transportation Research Part D: Transport and Environment*, *25*, 59–67.

Schweitzer, L., & Valenzuela Jr A. (2004). Environmental injustice and transportation: The claims and the evidence. *Journal of Planning Literature*, *18*(4), 383–398.

Spin. (2022). *Spin Access*. https://www.spin.app/spin-access

Taylor, B., & Morris, E. A. (2015). Public transportation objectives and rider demographics: Are transit's priorities poor public policy? *Transportation*, *42*(2), 347–367.

Taylor, B. D., & Norton, A. T. (2009). Paying for transportation what's a fair price? *Journal of Planning Literature*, *24*(1), 22–36.